भारतीय राष्ट्रीय भुगतान निगम
NATIONAL PAYMENTS CORPORATION OF INDIA

UNIFIED PAYMENTS INTERFACE

Procedural Guidelines

A procedural document that defines the procedural guidelines framed under the provisions of Payment and Settlement System Act, 2007 and are binding to all members of Unified Payments Interface

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Customer Registration Process:

UPI service should be provided to customers registered for mobile banking service if it is initiated from Mobile App. The PSP registration process should allow the customer to generate/obtain his virtual address with the PSP through the registration process. It is however, also possible for a Bank joining UPI only as an Issuer, to provide Virtual addresses to its customer base by default.

The customer shall be providing banking details to be mapped against this virtual address through the defined process. These fields available shall form the local mapper at the PSP end for which it may have the customer agree to specific 'Terms & Conditions'.

The User Interface (UI) guidelines are only for reference purposes. PSPs are free to innovate the user experience part. The guidelines related to steps with regard to customer registration on PSP App are outlined in *Annexure IV*.

Customer Complaints:

In case of any customer complaints regarding non refund for failed transactions and/or non-credit for successful transactions shall be dealt by the PSP/Bank. Any complaint about credit not being given to a beneficiary should be dealt with conclusively and bilaterally by the remitting and beneficiary banks as per the guidelines circulated by NPCI from time to time.

In case of any complaints related to UPI transactions, the first point of contact for customer will be the customer's PSP. Customer's PSP has to mandatorily provide option in their App to raise dispute/complaint by providing transaction reference/Id number. However, if customer decides to approach his/her remitter/beneficiary bank instead, the respective banks shall entertain all such requests and help to resolve the complaint to the customer's satisfaction. The PSP must provide to customers, the option of checking the current status of a transaction in the PSP App.

Mobile Banking Registration Transaction:

Mobile Banking registration transaction allows the customer to register for mobile banking service with his Issuer Bank through UPI. It will be possible only if the mobile number (which is to be registered) is registered with the Issuer Bank for SMS Alerts/mobile alerts & not for Mobile Banking services. This service won't allow the customer to change or modify any existing number. It will only elevate the mobile number registered for receiving alerts to full-fledged mobile banking services. The customer requires details like last six digits of debit card, expiry date and an OTP to authenticate for this transaction. This transaction is to be facilitated only through PSP App. The steps related to Mobile Banking registration, Set/Change PIN and Generate OTP through PSP App are outlined in *Annexure V*.

UPI Transaction Flows:

The UPI transactions flows are outlined in Annexure VI.

Roles & Responsibilities of the PSP:

The roles and responsibilities of the PSP are outlined in *Annexure - VII*

Roles & Responsibilities of the Sub-members:

The roles and responsibilities of the sub-members are outlined in *Annexure - VIII*

Broad Roles & Responsibilities of the Technology Service Provider (TSP):

The broad roles and responsibilities of the TSPs are outlined in *Annexure - IX*

UPI PSP role:

The UPI PSP roles are outlined in Annexure –XII

Broad Security Considerations:

The broad security considerations are outlined in Annexure – XIII

App Checklist:

The App checklist are outlined in Annexure - XIV

Annexure – I (Settlement & Reports)

UPI Settlement & Reports Availability:

The settlement of UPI transactions will be facilitated through IMPS. All the rules and regulations of IMPS Settlement will be applicable in UPI for transactions where the credit leg is processed through IMPS. No separate raw files, STL & VERAF will be available for such transactions. They will be part of existing IMPS raw files.

For transactions where both the debit and the credit leg of the transactions is processed through UPI, then separate raw files, DSR and other reports will be provided in the existing IMPS RGCS interface.

Following table provides the details

Sr.	Details	Debit & Credit (Customers)	Settlement
1	Sending Bank live in UPI & Recipient Bank live in IMPS only	Debit in UPI Credit in IMPS	Settlement in IMPS
2	Both the Sending & receiving Bank live in IMPS	Credit in IMPS	Settlement in IMPS
3	Both Banks live in UPI	Debit in UPI Credit in UPI	Settlement in IMPS

Note:

- The settlement cycles are subject to revisions from time to time.
- All UPI members should download the respective settlement files.
- Members must perform reconciliation as per the guidelines issued by NPCI from time to time.
- The Settlement cycles and timings shall be as prescribed by NPCI from time to time.
- UPI members should have a separate operations and reconciliation team to handle the day-to-day activities proactively and efficiently.

At the end of the each cutover time and completion of Settlement file generation, NDC limit would be refreshed, the net receivable or payable of each member would be generated, and a daily settlement report would be prepared and sent to all members through a Secure File Transfer Protocol (SFTP) and made available through the DMS/RGCS application.

Currently, NDC Limits are allotted to banks in IMPS as per the SGM policy. For UPI transactions, banks will be assigned a specific percentage of their existing IMPS NDC Limits as prescribed and as approved by the respective governance bodies.

Please note:

- The net settlement amount would include transaction and settlement fees payable between the banks & PSPs (Interchange Fee & PSP Fee) and NPCI (Switching Fee).

- NPCI has obtained Type D RTGS membership and provides settlement service to banks. It would be free to revise the settlement charges based on business needs.

- NPCI will act as a settlement agency and will arrange the necessary interbank settlement of credits and debits to the banks' respective RTGS Settlement Accounts with RBI as per approval received from RBI vide letter DAD/RTGS/626/24.02.001/2011-12 dated Oct 24, 2011.

- It will be the members' responsibility to verify accuracy of the Daily Settlement Reports with reference to the data available at their end.

- In case of net debit, a member has an obligation towards other members. Therefore, members are advised to ensure strict compliance to the RTGS operational instructions of RBI in this connection. Any failure to maintain the required balance in the RTGS settlement account would attract stringent action as deemed fit by NPCI.

- Letter of authority: For existing banks which are live on IMPS, existing mandate to RBI for credit/debit of settlement account will suffice.

- New members before participating in UPI should issue letter of authority to RBI authorizing net debit/credit for UPI related transaction in their respective RBI accounts by NPCI, duly approved by their respective boards. This is applicable to the members who are not live on IMPS.

- A member failing to meet its daily settlement obligation more than two times in a month would be debarred from membership.

Any changes due to settlement of UPI transactions in IMPS Settlement shall be made at later stage if required.

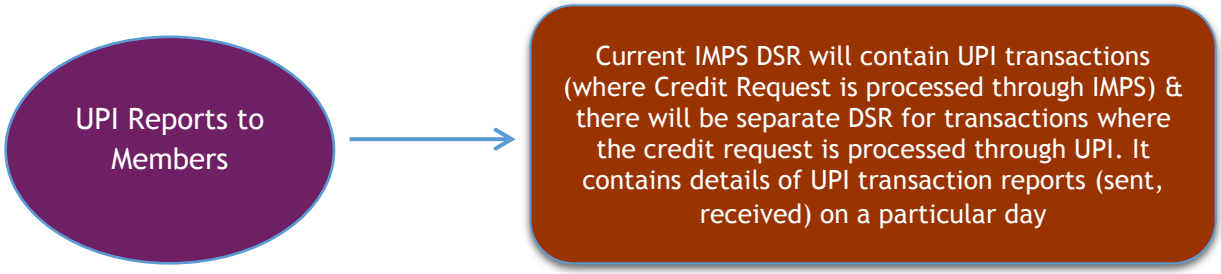
Reports Availability:

UPI would provide the following daily reports in DMS / RGCS application format round-the-clock:

- Raw data file
- Net settlement report (NTSL)/Daily Settlement Report (DSR)*
- Settlement file (STL) – on NPCI settlement day
- Any other reports as may be relevant

* Available only during RTGS working days for settlement

Members will be provided with the following reports:



Adjustment to Settlement:

Discrepancies relating to reconciliation/adjustment done by members, based on reports furnished by UPI are the responsibility of the participating members. Such discrepancies should be resolved by members as per the settlement procedures set forth in the UPI Procedural Guidelines. The following points explain the switching fee adjustments:

- NPCI determines the amount of service fees its members owe for using UPI services.
- As a service provider, NPCI would maintain an account with a member participating in the UPI network.

Dispute Management:

There will be broadly two types of variants in UPI Disputes Management & Settlement:

a) U2U

- Denotes transactions where both the debit and credit request are processed through UPI and will be available in separate raw data, reports, settlement files & DSR.
- In case of timed-out transactions (cases where NPCI was not able to receive the response in a specified time frame from Remitter/Beneficiary Bank), settlement of such cases will not be performed as this will be considered as declined transactions. The debit and the credit reversals will be processed.
- Accordingly, all the transactions will be categorised as either a successful or a failed transactions and there will not be any transactions with pending status.

Timeouts:

Timeouts in UPI will treated as declined transaction, wherein Reversal will be generated by Beneficiary bank and/or NPCI as applicable:

- **Wrong Credits / Closed / Frozen Accounts:**
  Credit adjustment option has been given to beneficiary banks. Beneficiary banks will raise credit adjustment once they recover funds successfully from their customer.
- **Complaint Window to PSPs:**
  PSPs (who are not remitter bank or not a beneficiary bank for particular transactions) are also allowed to raise complaint on either remitting bank or beneficiary bank for UPI transactions. E.g.: If online reversal fails and customer a/c is still in debit.
- **Search Transactions & Adjustment Status:**
  All parties of the UPI will be provided to search Transaction & Adjustment status through RGCS system.
- **UPI Merchant Payment Transactions:**
  For merchant Payments transactions, existing chargeback cycle will be applicable.

b) U2I:

- Denotes transactions the debit request is processed through UPI and the credit request is processed through IMPS and will be available in existing IMPS raw data and other related files.
- In case the credit leg is processed through IMPS, then the existing rules of IMPS Settlement will be applicable for such transactions.

Both U2U & U2I transactions will be posted as a single entry as IMPS Settlement entry, but banks will be able to differentiate between the same as separate reports will be provided.

NPCI will issue circular from time to time in case of any changes in the Dispute Mechanism.

For Detailed dispute management processes, please refer to the UPI Operating and settlement guidelines.

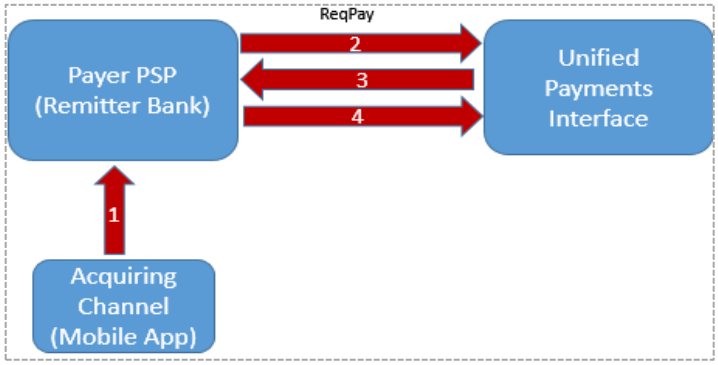
Annexure -VI (Flows of Financial Transactions)

PROCESS FLOWS FOR FINANCIAL TRANSACTIONS:

A. Transaction Flow when NPCI Libraries are used:

For Pay requests & Collect Request initiated/authorised from Payer PSP App which is different from the Remitter Bank, NPCI libraries will be used for capturing the auth credentials.

1. Transaction is initiated from PSP App. Customer enters the relevant details and authorizes by entering the authorization credential (PIN)
2. Payer PSP encrypts it using NPCI Public Key libraries & sends it to UPI
3. UPI decrypts it using NPCI Private Key sends it to the Remitter Bank by encrypting it with Bank's Public Key
4. Remitter Bank decrypts it using its Private Key & validates it, debits the customer's account and responds to UPI

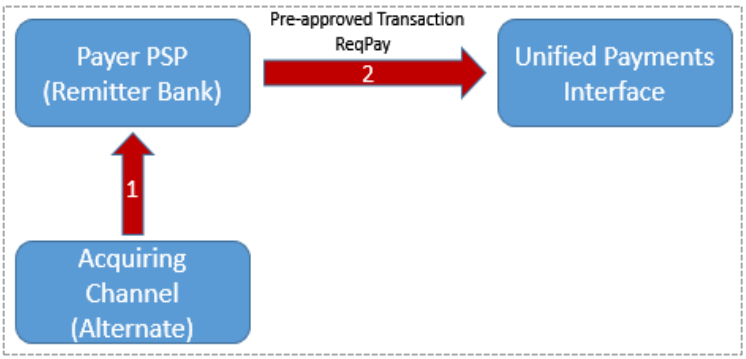


For illustration purposes, Step 3 & 4 is not provided in the process flows for non pre-approved transactions

B. Transaction Flow when NPCI Libraries are not used::

For transactions initiated from alternate channels, NPCI libraries will not be used for capturing the auth credentials and the bank can send it as pre-approved transactions where the role of UPI wil be to process the Credit request. It can be initiated in cases where the Payer PSP &Remitter Bank are same entity. Below is the sample process flow.

1. Transaction is initiated from PSP App. Customer enters the relevant details and authorizes the transaction. Remitter Bank debits the customer's account.
2. Payer PSP sends this transaction to UPI for processing the credit request.



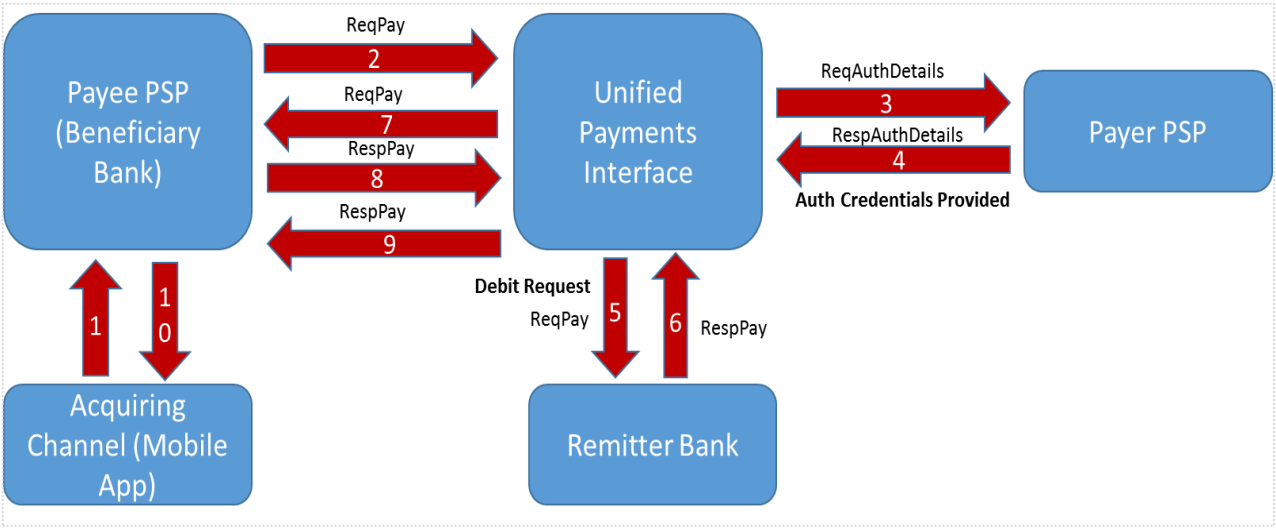
C. Push Transactions with Addresses like Aadhaar Number:

Global Addresses are Aadhaar Number. These global identifiers will be mapped to Banks in the NPCI Mapper. The data will be provided by the banks.

Three Party Model

(Pull : Remitter Bank & Payer PSP are separate entities, Payee Bank & Beneficiary Bank are one entity entity)

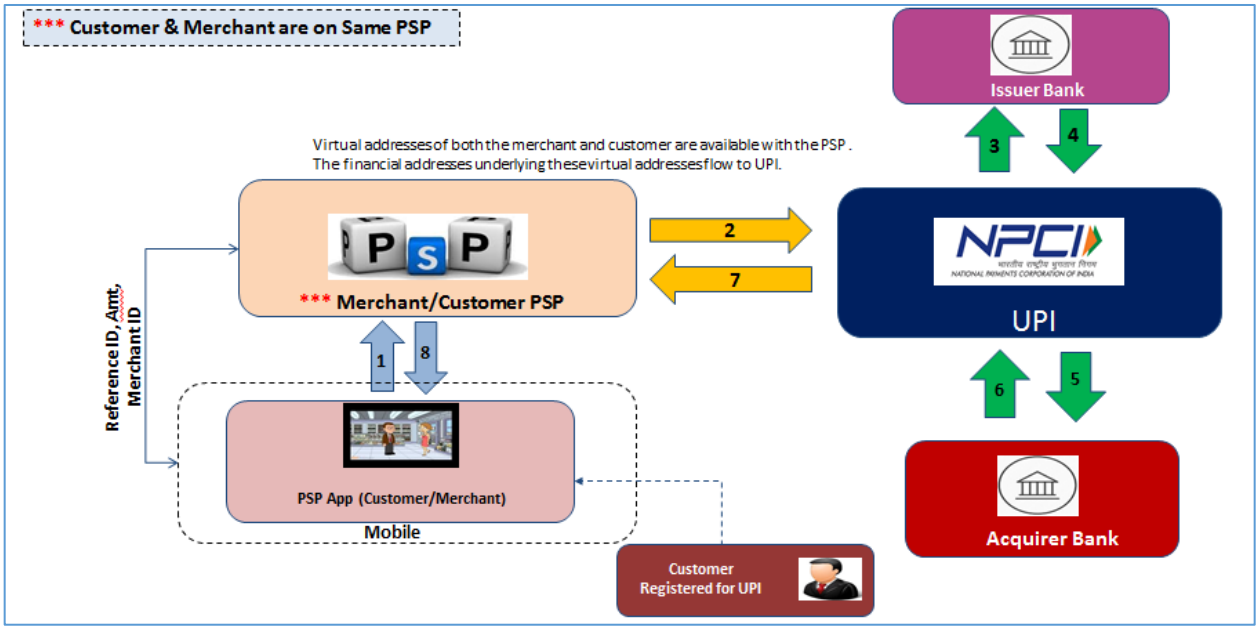
1. Customer sends a Collect Request by entering the Virtual Address of the Payer customer.
2. Payee PSP sends the same to UPI
3. UPI sends it to the respective Payer PSP for address resolution and authorization
4. Payer PSP sends a notification to the Payer customer for authorization. Customer enters the PIN & confirms the payment. Payer PSP sends the same to UPI
5. UPI sends the debit request to Payer bank.
6. Remitter bank debits the Payer's account and sends the confirmation to UPI.
7. UPI sends the credit request to the Beneficiary Bank
8. Beneficiary Bank credits the customer's account and confirms the same to UPI
9. UPI sends the successful confirmation to the Payee PSP
10. Payee PSP sends the confirmation to the customer



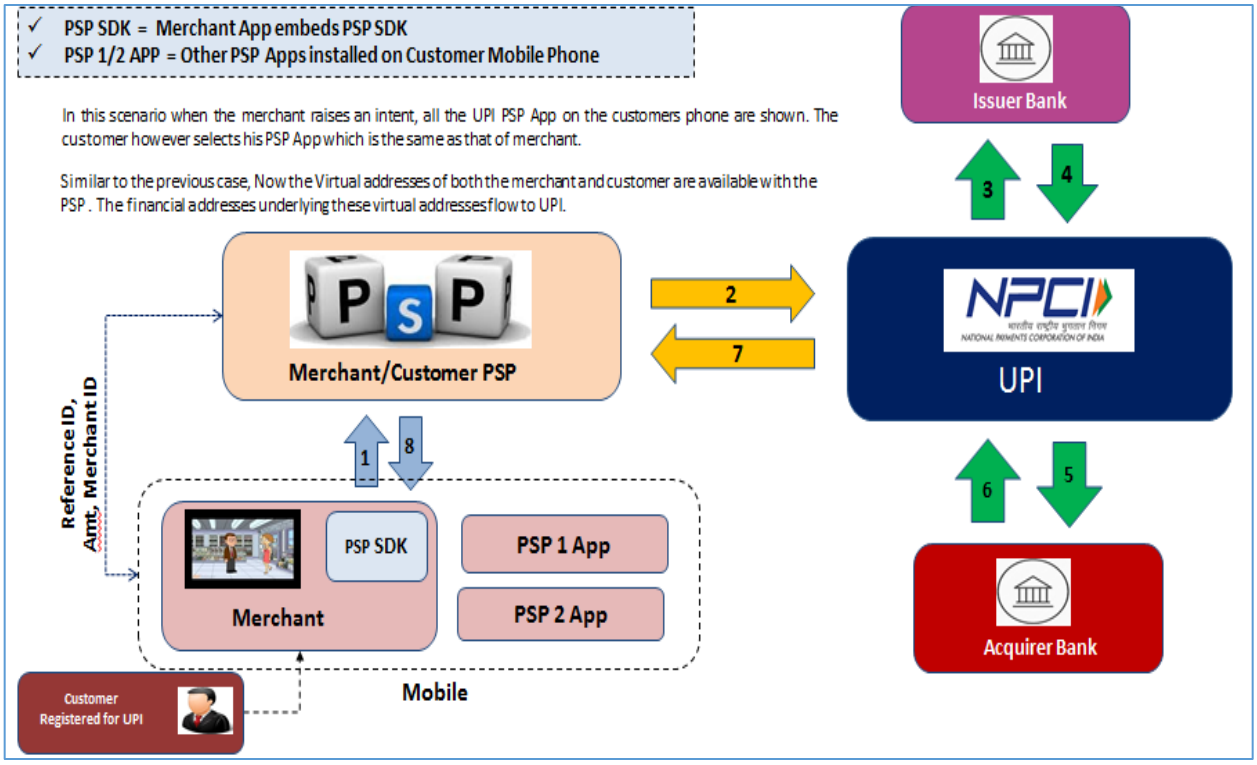
Three Party Model

(Pull : Remitter Bank & Payer PSP are one entity, Payee Bank & Beneficiary Bank are separate entities)

1. Customer sends a Collect Request by entering the Virtual Address of the Payer customer.
2. Payee PSP sends the same to UPI
3. UPI sends it to the respective Payer PSP for address resolution and authorization
4. Payer PSP sends a notification to the Payer customer for authorization. Customer enters the PIN & confirms the payment. Remitter Bank debits customer account and sends response to UPI.
5. UPI sends the credit request to the Beneficiary Bank
6. Beneficiary Bank credits the customer's account and confirms the same to UPI
7. UPI sends the successful confirmation to the Payee PSP



Case 2 : 3 Party Merchant Transaction - Customer & Merchant with Same PSP and PSP SDK embedded into Merchant App



Annexure XIII (BROAD SECURITY CONSIDERATIONS)

1. **Saving the Collect requests (Marked Safe) received towards future authentications:**

   a) UPI Pull functionality provides for a Collect request of money originating from the end beneficiary.
   b) An individual's smart phone gets a Push notification from its PSP that a request for seeking /collecting money has been initiated by the end beneficiary.
   c) The message along with the virtual address of the beneficiary for e.g. rachitsoral@abcbank would also have the name being displayed on the App i.e. "Rachit Soral". Mr. Rachit Soral being an acquaintance of the sender, the sender knows that it is safe to send money for the request that has been originated and can treat the person as a white-listed entity/person for future requests. This can be made possible through population of the values in the element 'Verified Name' in the Payer / Payee tags. The name is as per the details stored in the CBS of the bank and is picked up without any customer intervention.
   d) The Recipient should be able to save such Collect Requests as references for future and mark these incoming requests as "Safe" for all future Collect requests coming in from this particular sender.

This shall ensure that scrupulous and Spam collect calls can be prevented where the beneficiaries are already known and registered through the "Save" option by the recipient.

2. **Display of the Name on the PSP Mobile App while adding the Account (Pulled from CBS) / Name display on the remittance transaction either Collect OR Pull requests**

   a) While the customer is adding the Account to the Profile / Virtual address, he/she selects the Bank name for which the Bank account has to be added. The request for account fetching goes from the PSP App to UPI to the Issuing Bank.
   b) The PSP App should also display the Account Name of the customer as it is stored in the Core banking system of the Bank. This will provide the actual name to be stored in the App since it is completely automated with NIL customer intervention.
   c) Like-wise the name of the customer should be displayed on the App – both in the case of sending monies (who is the sender) and for Collecting money (who is the initiator of the Collect request).

These steps would help ensure that any fraud pertaining to the wrong representation by the perpetrator can be prevented.

3. **Merchant white listing in UPI**

Under UPI, it is possible to initiate Merchant transactions under the following 3 modes/methods:

   a) Collect request originated over the Web App or Internet App by the merchant: This payment is initiated when the customer has selected the payment option as "Pay by UPI". This would send a Push notification to the customer's smartphone through his PSP App and the customer can enter his Mpin and initiate a Debit on the account.

   b) The Push transaction – Where-in the customer knows the Virtual address and/or IFSC and Account number of the Merchant & he can Debit his account to pay the Merchant.

c) In – App Payment method : Where-in the Merchant's Mobile App 'calls/invokes' all the UPI certified PSP Apps on the customer's phone & the customer selects his/her preferred App for making the payment by entering his MPIN on the PSP App.

It is recommended by NPCI that the large merchants be white listed by the PSP Bank. The merchant can be "whitelisted" by the PSP Bank using APIs (CreateWLentry / UpdateWLentry/GetWLentry), checking with the centrally stored merchant details. This will help ensure that the customer has confidence in terms of the Merchant relationship with the PSP bank along with credibility and he/she can initiate the payment seamlessly. The large merchants would ideally include the likes of "IRCTC", "Flipkart", "Amazon", "Snapdeal" and "BookMyShow" etc.

NPCI shall incrementally provide the list of large merchants enabled on UPI to the PSPs on a monthly basis. These details can also be marked at the PSP server level basis the information provided by NPCI.

(Note: Names have been used only for representation purpose)

4. Name Check for Virtual addresses created by customer

a) This implies holding a list of such names at the PSP Server which cannot be added.
b) The list of top 100/200 names can be stored in the database by the PSP, against which the names being entered by the customers shall be checked. If the name being entered in the list falls in the negative list, it will not be allowed. (NPCI shall endeavor to provide such list of names to the member banks)
c) Further, the display of names as in CBS under the 'Verified Name" element in the Payer/Payee tags ensures, that – even though the requesting Virtual Address may read as ***presidentofindia@abcbank*****,** the name of individual actually originating the request, shall be shown as 'Mr. Rachit Soral' and not 'Shri Pranab Mukherjee'.
d) Further, once the Virtual Address has been opted by a customer and later the profile is closed, then the PSP should not allow this VPA to be used by any other person/entity at least till 2 years post the deactivation by the initial customer.

The display of name in the collect request of the initiating customer is therefore a must for the PSP Banks. The name can be picked up from the UPI online message as stated above.

5. Mask Collect Request - check at NPCI to decline the request

a) It also needs to be ensured that the Collect requests received by the Beneficiary are not in a masked format. If this were to be permissible, then the requirement of displaying the name of the initiator of collect request is defeated.
b) The name should appear in clear on the PSP App for the customer to see. NPCI shall put in a process of terminating a request that is received in the message where the name and/or any other required details are in a masked format.

This would be required from the perspective of preventing any fraudulent and spam requests being received by the payer.

6. Velocity check of multiple credits to single account at beneficiary Bank end

a) The Collect requests can be initiated by an individual registered for the PSP App of any bank on any other individual who has registered and has created a Virtual address at the time of registration of the User profile with the UPI App.

b) Basis the Virtual Address, It is possible for an initiator to send multiple Collect requests to a single Virtual address such as Mr. Vikrant Sharma (vikrant@xyzbank) sending multiple collect requests on the virtual address rachitsoral@abcbank.
c) This is an avoidable nuisance for the recipient if he/she is flooded with such requests on a continual basis and are akin to Spam requests.

Recommendation would be have a velocity check on the following parameters:

1) Not more than 10 transactions in a day on the same VPA
2) Not more than 25 transactions in a week on the same VPA.

In such cases the following are recommended:

a) For Individuals: As a Beneficiary PSP, the PSP Bank should build in checks to ascertain the number of such requests coming in from a particular virtual address (viz - vikrant@xyzbank) and if such requests exceed the prescribed threshold set by the PSP for an individual V.A sending collect request, then action could be taken at the PSP level to block any more requests coming in for its customer OR may take any such other action as is deemed appropriate.

b) For Merchants: This step mentioned above can also be utilized by the Acquiring PSP Bank for a merchant transaction. It is possible within the UPI framework for a Merchant to initiate a Collect request from the end customer as an individual (P2P) payment, whereas it is actually a merchant transaction. A merchant may resort to such means in order to escape the MDR/Interchange applicable in the Merchant transaction cycle.

a) It is therefore critical that the Beneficiary PSP Bank/Acquiring Bank in this case have a check on the velocity of such transactions coming in to the same Virtual Address.
b) It is an indication for the PSP Bank that excessive collect requests for a particular Virtual address may actually be a merchant transaction where the entity is collecting money on a P2P basis. PSP Bank could then initiate an inquiry / assessment, basis which the individual should then be set up as a merchant in the UPI framework along with the respective MCC and the corresponding MDR/Interchange. NPCI has also prescribed Merchant Compliance guidelines in this regard.

Note: NPCI shall provide report on velocity of the transactions of merchants towards aiding appropriate decision making by the PSP.

7. Prudent steps to be followed where the Mobile number / Mobile device has been changed.

7. a) Change in the Mobile Device:

a) In case of the change in the handset, it is mandatory basis the UPI framework to download the App again.
b) In this case, the mobile number shall again send the encrypted SMS to the PSP Interface. The PSP interface identifies that the same mobile number had been previously registered with it.
c) The PSP App should allow the customer to continue with the existing Virtual address, while it creates the new Device Hard-biding basis the encrypted SMS sent to the PSP interface.

7. b) Change in the mobile number:

a) A change in the mobile number would require the customer to give the new number / Update the number with his/her Issuing Bank.

b) The customer would be required to download the UPI App again using the same mobile number.
c) The PSP app sends the encrypted SMS to the PSP interface for creating the Device Hard binding.
d) The customer would be required to create a new Virtual Address to continue & this will be treated as a new registration by the PSP. This is to prevent any fraud related to usage of VPA.

Note: If the PSP can assess the veracity of the customer basis some secret Question / Answers, details of which were captured at the time of initial registration – the PSP may permit the customer to use the same VPA in-spite of he/she using a new mobile number.

8. Outbound encrypted SMS

a) The most critical security requirement is to bind the mobile number with the device at the time of customer downloading the PSP App and creating a profile.
b) It has been mandated that the PSP App shall send an outward encrypted message to the PSP interface from the mobile number being used by the customer for registering the PSP profile. It is also recommended that this SMS is sent for every critical change (for e.g. upgrade of the OS version on the handset) to maintain security level checks).
c) In doing so, the PSP interface creates a Device fingerprinting of the mobile number, closely bound with Device Id, App Id, IMEI number or any other.
d) The outward SMS being sent is initiated automatically from the PSP App invoking the mobile number. There is no intervention of the customer to send this message out.
e) However, an alert may be generated by the PSP App indicating to the customer that an outward message is being initiated which would cost a nominal 'SMS charge" to the customer.
f) It is this device "hard biding" that shall be used by the PSP for authenticating the first factor of the subsequent financial transaction.
g) Secondly, the Mobile number being bound to the App, the hard bound mobile number (authenticated by the PSP) also becomes the carrier of the information in an interoperable transaction bringing in the trust between the PSP & the Issuer.
h) **Rooted Devices (Android) / Jailbreaking (iOS**) – for the rooted devices where the customer has control, the PSP may decide whether it wants to let the customer continue towards creating of the profile (for security reasons). The PSPs may have internal policies in this regard.

Annexure XIV (APP CHECKLIST)

The purpose of the below Checklist is to ensure that the UPI based Application have the same look and feel across applications of different entities. The Field Marked with * has to be filled by NPCI official only after ensuring sanctity of product.

Note: -
1. Application release in public domain cannot happen without this Checklist clearance.
2. 'M' implies Mandatory functionality
3. 'O' implies Optional functionality

Level 1:- How to Download an UPI enabled Application.

Sr.	Location	Bank/PSP Name in the App name	Mandatory* /Optional	Yes / No*
a	Play Store		M	
B	App Store		M	
C	Windows Store		O	

* - As per the UPI PG

Level 2:- Home screen checklist

Sr.	Transaction Type		Mandatory / Optional	Yes /No*
a	Send(Push)	1. Virtual Address (VPA) 2. Account /IFSC 3. Mobile No/MMID	M M O O	
b	Collect	1. Virtual Address 2. Aadhaar	M O	
c	Add/Link a Bank A/C		M	
d	Transaction history		M	
e	Mobile Banking registration	1. Card No. & Expiry is manual entry. 2. Picture/Scan	M O	
f	Generate Mpin/ Set Mpin	1. Card No. & Expiry is manual entry. 2. Picture/Scan	M O	
g	Change Mpin	Old mPIN & new mPIN is manual entry	M	
h	Change Application Password		O	
i	Log a complaint		M	

Level 3:- How to register and deregister on PSP application.

Sr.	Action	Mandatory/Optional	Yes /No*
a	Encrypted SMS from App without User	**M**	
a.1	Type of Encryption	**M**	**PKI**
a.2	Choice of SIM Selection should be there in case of Dual SIM phones.	**M**	
b	Profile Creation	**M**	
c	VPA Creation	**M**	
d	Application password	**O**	
e	Login Page to restart application	**O**	
f	Deregistration from PSP	**M**	
g	Deletion of Account linked to VPA	**M**	
h	Deletion of VPA	**M**	

Level 4:- Total User entry

Sr.	Action	Mandatory/Optional	Yes /No*
a	Entry of Mobile No	**M**	**No**
b	Virtual Payment Address	**M**	**Yes**
c	Debit Card entry /Scan	**O**	**Manual /Scan**
d	SET MPIN	**M**	**Yes/No**
e	Whitelisting of Collect Requestor **(good**	**O**	**Yes/No**
f	Whitelisting of Merchant **(good to have)**	**O**	**Yes/No**
g	Storing of Beneficiary by nick name for following • VPA • IFS + A/C No • Aadhaar	**O**	**Yes/No**
h	Having the Favorite Selection of Beneficiary	**O**	**Yes/No**
i	Restoring of VPA by PSP in case of deletion of VPA by user	**M - at least 2 years Moratorium**	**Yes/No**

Level 5:- Add a bank Account / VPA & Account handling

Sr.	Actions	Mandatory/Optional	Yes / No*
a	List of bank to be displayed as an Issuer in drop down menu	**M**	
b	List of bank to be displayed as an Issuer in drop down menu to be search by start letter of the bank	**M**	
c	If customer registered for MB display Set Mpin transaction based on Debit card last 6 digit & expiry date followed by OTP from Issuer **OR** If customer has MPIN, app should provide option – "Continue with Existing MPIN".	**M**	
d	If customer not registered for MB customer should be routed to MBR transaction with the same details required for Set MPIN	**M**	
e	One VPA to multiple Account (Default VPA selection) - **Optional**	**O**	
f	Multiple VPA to Single Account	**O**	
G	One PSP APP to register Multiple bank account with multiple VPAs	**M**	
h	Generate OTP	**O**	
h.1	OTP read by the App OR entered Manually	**O**	

Level 6:- Pay Transaction

Sr.	Actions	Mandatory/Optional	Yes/ No*	Remarks
a	Pay using VPA	**M**		
b	Pay using a/c & IFSC	**M**		
c	Pay using mob & MMID	**O**		
d	Pay using Aadhaar	**O**		
e	Mpin Preapproved	**M**		
f	Mpin non pre-approved	**O**		
g	Transaction online Confirmation	**M**		
h	Methodology of invoking Common Library • Mpin preapproved / onus transaction invocation of common library • Mpin non-preapproved / off us transaction	**O** **M**		

	invocation of common library. • Ref URL in common library on clicking should take customer to relevant page with complete detail (For example Bill no, amount, transactions id and complete bill details).	**M**		

Level 7:- Collect transaction

Sr.	Actions	Mandatory/Optional	Yes/ No*	Remarks
a	Collect using VPA	**M**		
b	Mpin Preapproved	**M**		
c	Mpin non preapproved	**O**		
d	Transaction Status Confirmation	**M**		
e	Display of VPA and Name for incoming collect request	**M**		
f	Display of Expiry time in case of incoming collect request	**M**		
g	Default validity of 30 minutes in case customer is not specifying the expiry time	**M**		
h	Minimum validity of 1 Minutes in case customer is selecting expiry time of collect request explicitly	**M**		
i	Methodology of invoking Common Library • Mpin preapproved / onus transaction invocation of common library • Mpin non-preapproved / off us transaction invocation of common library. • Ref URL in common library on clicking should take customer to relevant page with complete detail (For example Bill no, amount,	**O** **M** **M**		

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	transactions id and complete bill details).			

Level 8:- Balance Enquiry

Sr.	Action	Mandatory/Optional	Yes/ No*	Remarks (If Any)
a	Select Account	**M**		
b	Mpin Preapproved	**M**		
C	Mpin Non Preapproved	**M**		

Level 9:- Check Transaction Status / Raise Query/ log a complaint

Sr.	Action		Yes /No *	Remarks (if Any)
a	Transaction History • Ref no. wise • Date wise	**O** **M**		**Banks to decide on the parameters**
b	Raise a query / Dispute • Basis On ref no. • date wise	**O** **M**		**Banks to decide on the parameters**
c	While raising the query/dispute, the App should display • Transaction ID • Beneficiary and Remitter VPA/other address • Date & Time • Amount	**M**		
d	Check Transaction Status • Option Raising a query/ Log a complaint against each transaction should be there.	**M**		
e	Last 5 Transactions	**M**		

Level 10:- Hot listing/Deletion

Sr.	Action	Mandatory/Optional	Yes/ No*	Remarks (If Any)
a	Once hot listed or deleted, is Bank allowing to re-allocate the same VPA?	**M (2 years Moratorium)**		

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Level 11:- Payments by UPI

Sr.	Action	Mandatory/Optional	Yes/ No*	Remarks (If Any)
a	Pay by UPI	**M**		**Nomenclature is to be decided by the bank**
b	Collect by UPI	**M**		**Nomenclature is to be decided by the bank**
C	Reject Collect request	**M**		**Nomenclature is to be decided by the bank**

Level 12:- QR Code based enablement

Sr.	Action	Mandatory/Optional	Yes/ No*	Remarks (If Any)
a	Customer being able to generate the QR Code within his App (Which Merchant can scan for payment)	**O**		**Merchant based Pull Payments**
b	Customer should be able to scan the QR Code of a Merchant / Another Person for making payments	**O**		**Push payment - Merchant / P2P**
c	It is mandatory to display to the customer at least VPA, Amount and name in QR based payment.	**M**		**Push payment - Merchant / P2P and Merchant based Pull Payments**

Level 13:- Raising Intent Call

Sr.	Action	Mandatory/Optional	Yes/ No*
A	The App to support Intent Call request from the Merchant Apps		
a.1	Android	**M**	
a.2	IOS	**M**	
a.3	Windows	**M**	

Additional Comments (if any):-

Sign off

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